

# Kristina Monakhova

ASSISTANT PROFESSOR @ CORNELL CS

✉ monakhova@cornell.edu | 🏠 kristinamonakhova.com

## Appointments

---

### Cornell University

*Ithaca, NY*

ASSISTANT PROFESSOR, DEPARTMENT OF COMPUTER SCIENCE

8/1/2024 - present

- Leading the Cornell Computational Imaging Lab
- Field member of Electrical and Computer Engineering, and Applied Mathematics

### Massachusetts Institute of Technology

*Cambridge, MA*

POSTDOCTORAL FELLOW

9/2022-8/24

Working with Sixian You and George Barbastathis

## Education

---

### University of California, Berkeley

*Berkeley, CA*

PH.D. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCES

Aug. 2022

- Advisor: Prof. Laura Waller
- Dissertation: Physics-Informed Machine Learning for Computational Imaging [pdf]

### The State University of New York at Buffalo

*Buffalo, NY*

BS, ELECTRICAL ENGINEERING

May 2016

## Academic Honors & Awards

---

MIT Postdoctoral Fellowship for Engineering Excellence

2022

UC Berkeley EECS Demetri Angelakos Memorial Achievement, for department service and altruism

2021

UC Berkeley EECS Chairs' Graduate Award, for organizing EE Visit Days

2020

UC Berkeley EECS Excellence Award

2016

National Science Foundation Graduate Research Fellowship (NSF GRFP)

2016

National Defense Science and Engineering Graduate Fellowship (NDSEG), (declined)

2016

Barry M. Goldwater Scholarship

2015

University at Buffalo Presidential Scholarship, four year full-ride scholarship

2012 – 2016

## Previous Professional Experience

---

### Intel Intelligent Systems Lab

*Santa Clara, CA*

GRADUATE RESEARCH INTERN

Spring 2021

- with Dr. Vladlen Koltun

### MIT Lincoln Laboratory

*Lincoln, MA*

ADVANCED SENSOR SYSTEMS AND TEST BEDS INTERN

summer 2016

## Carnegie Mellon University Robotics Institute

RISS REU UNDERGRADUATE RESEARCHER

- with Dr. Red Whittaker

Pittsburgh, PA

summer 2014

## NASA Marshall Space Flight Center

NASA ROBOTICS ACADEMY SUMMER RESEARCHER

Huntsville, AL

summer 2013

## Teaching

---

### INSTRUCTOR, CORNELL UNIVERSITY

CS 4660 - Foundations of Computational Imaging

Spring 2026

CS 6662 - Computational Imaging

Fall 2024

### GRADUATE TEACHING ASSISTANT, UC BERKELEY

EE16A - Designing Information Devices and Systems I

Fall 2020

EE16A - Designing Information Devices and Systems I

Summer 2020

EE290T - High dimensional data analysis with low dimensional models

Fall 2018

### GUEST LECTURES

UCSD ECE 285: Computational Imaging Systems

Spring 2024, 2025

Lecture title: 'Robust and physics-informed machine learning for low light imaging'

MIT 6.300: Signal Processing

Spring 2024

Lecture title: 'Computational Imaging: using signal processing to make better imaging systems'

## Publications

---

\*indicates equal contribution

† indicates undergraduate researcher under my supervision

### JOURNAL AND JOURNAL EQUIVALENT PUBLICATIONS

1. Christian Foley, Eric Markley, Kyrolos Yanny, Laura Waller, **Kristina Monakhova**, "Spectral DefocusCam: super-resolved hyperspectral imaging through defocus", ICCP 2025. [pdf]
2. Xiang Dai, Kyrolos Yanny, **Kristina Monakhova**, Nicholas Antipa, "Single-shot HDR using conventional image sensor shutter functions and optical randomization," ACM Trans. Graph., July 2025. [pdf]
3. Jiashu Han, Kunzan Liu, Keith B Isaacson, **Kristina Monakhova**, Linda G Griffith, Sixian You, "System- and sample-agnostic isotropic three-dimensional microscopy by weakly physics-informed, domain-shift-resistant axial deblurring," Nat Commun 16, 745 (2025). [pdf]
4. Cassandra Tong Ye, Jiashu Han, Kunzan Liu, Anastasios Angelopoulos, Linda Griffith, **Kristina Monakhova**, and Sixian You, "Learned, uncertainty-driven adaptive acquisition for photon-efficient scanning microscopy," Opt. Express 33, 12269-12287 (2025) [pdf]
5. Michael Hecht, Artur Yakimovich, **Kristina Monakhova**, Laura Waller, et al., "Roadmap on Machine Learning for Microscopy", (in review) [pdf]
6. **Kristina Monakhova**, Stephan Richter, Laura Waller, Vladlen Koltun, "Dancing under the stars: video denoising in starlight," CVPR 2022 (Oral, 4% acceptance rate), [pdf]
7. Kyrolos Yanny\*, **Kristina Monakhova**\*, Richard W. Shuai, Laura Waller, "Deep learning for fast spatially-varying deconvolution," Optica, 9 (1), 2022 [pdf]

8. **Kristina Monakhova\***, Vi Tran<sup>\*†</sup>, Grace Kuo, Laura Waller, “Untrained networks for compressive lensless photography,” *Opt. Express* 29, 20913-20929 (2021) [pdf]
9. **Kristina Monakhova\***, Kyrollos Yanny\*, Neerja Aggarwal, Laura Waller, “Spectral DiffuserCam: lensless snapshot hyperspectral imaging with a spectral filter array,” *Optica*, 7 (10), pp. 1298–1307, 2020 [pdf]
10. Kyrollos Yanny\*, Nick Antipa\*, William Liberti, Sam Dehaeck, **Kristina Monakhova**, Fanglin Lina Liu, Konlin Shen, Ren Ng, and Laura Waller, “Miniscope3D: optimized single-shot miniature 3D fluorescence microscopy,” *Light: Science & Applications*, 9 (171), 2020 [pdf]
11. **Kristina Monakhova**, Joshua Yurtsever<sup>†</sup>, Grace Kuo, Nick Antipa, Kyrollos Yanny, and Laura Waller, “Learned reconstructions for practical mask-based lensless imaging,” *Opt. Express* 27, 28075-28090 (2019) [pdf]

## CONFERENCE PUBLICATIONS

1. Cassandra Tong Ye, Jiashu Han, Kunzan Liu, Anastasios Angelopoulos, Linda Griffith, **Kristina Monakhova**, Sixian You “Leveraging uncertainty quantification in adaptive multiphoton microscopy acquisition”, *SPIE Computational Optical Imaging and Artificial Intelligence in Biomedical Sciences*, 2023. [pdf]
2. Christian Foley<sup>†</sup>, **Kristina Monakhova**, Kyrollos Yanny, Laura Waller, “Spectral DefocusCam: Hyperspectral Imaging Using Defocus and A Spectral Filter Array,” *Imaging and Applied Optics Congress, CF2C*, Optical Society of America, 2022. [pdf]
3. Neerja Aggarwal, Eric Markley, **Kristina Monakhova**, Kyrollos Yanny, Laura Waller, “Compact snapshot hyperspectral imager for fluorescence microscopy,” *Focus on Microscopy*, 2022 [pdf]
4. Richard W. Shuai\*, Kyrollos Yanny\*, **Kristina Monakhova**, Laura Waller, “MultiWienerNet: Deep Learning for Fast Shift-Varying Deconvolution,” *Imaging and Applied Optics Congress, CTh5A.5*, Optical Society of America, 2021.[pdf]
5. **Kristina Monakhova\***, Kyrollos Yanny\*, and Laura Waller, “Snapshot hyperspectral imaging using a random phase mask and spectral filter array,” *Imaging and Applied Optics Congress*, pp. JF2F.4, Optical Society of America, 2020. [pdf]
6. Grace Kuo, **Kristina Monakhova**, Kyrollos Yanny, Ren Ng, and Laura Waller, “Spatially-varying microscope calibration from unstructured sparse inputs,” *Imaging and Applied Optics Congress*, pp. CF4C.4, Optical Society of America, 2020. [pdf]
7. Ellin Zhao<sup>†</sup>, Nicolas Deshler<sup>†</sup>, **Kristina Monakhova**, Laura Waller, “Multi-sensor lensless imaging: synthetic large-format sensing with a disjoint sensor array.” *Imaging and Applied Optics Congress*, pp. CF2C.6, Optical Society of America, 2020. [pdf]
8. Kyrollos Yanny, Nick Antipa, William Liberti, Sam Dehaeck, **Kristina Monakhova**, Fanglin Lina Liu, Konlin Shen, Ren Ng, and Laura Waller, “Compressed Sensing Mask-based Miniature 3D Fluorescence Microscopy” *Imaging and Applied Optics Congress*, pp. CW4B.5, Optical Society of America, 2020. [pdf]
9. **Kristina Monakhova**, Nick Antipa, and Laura Waller, “Learning for lensless mask-based imaging,” in *Computational Optical Sensing and Imaging*, pp. CTu3A–2, Optical Society of America, 2019 [pdf]

## WORKSHOPS AND POSTERS

1. Hasindu Kariyawasam, **Kristina Monakhova**, “Deep Optics Design for Two-Shot Depth Estimation” in *International Conference on Computational Photography*, Toronto, July 2025
2. Cassandra Tong Ye, Shamus Li, Tyler King, **Kristina Monakhova**, “QUTCC: Quantile Uncertainty Training and Conformal Calibration for Imaging Inverse Problems” in *International Conference on Computational Photography*, Toronto, July 2025 (spotlight)
3. Shamus Li, Vi Tran, Ruiming Cao, Laura Waller, **Kristina Monakhova**, Sara Fridovich-Keil, “Single-Shot Novel View Synthesis with a Multiplexed Light Field Camera” in *International Conference on Computational Photography*, Toronto, Canada, July 2025 (spotlight)
4. Cassandra Tong Ye, Tyler King, **Kristina Monakhova**. “Quantile UQNet: A Conformal, Nonlinear Scaling Model for Uncertainty Prediction” in *Workshop on Uncertainty Quantification for Computer Vision, CVPR 2025*, June 2025 (spotlight talk)

5. **Kristina Monakhova\***, Vi Tran<sup>\*†</sup>, Grace Kuo, Laura Waller, “Untrained networks for compressive lensless photography” in CVPR Computational Cameras and Displays (CCD) Workshop, June 2021 (spotlight talk)
6. **Kristina Monakhova\***, Kyrollos Yanny\*, Neerja Aggarwal, Laura Waller, “Spectral DiffuserCam: lensless snapshot hyperspectral imaging with a spectral filter array,” in CVPR Computational Cameras and Displays (CCD) Workshop, June 2020 (spotlight talk)
7. Grace Kuo, Fanglin (Linda) Liu, **Kristina Monakhova**, Kyrollos Yanny, Ren Ng, Laura Waller, “On-chip fluorescence microscopy with a random microlens diffuser”, in 2020 ICCP Conference, St. Louis, MO, Apr. 2020 (poster)
8. **Kristina Monakhova**, Joshua Yurtsever<sup>†</sup>, Grace Kuo, Nick Antipa, Kyrollos Yanny, Laura Waller, “Unrolled, model-based networks for lensless imaging”, 2019 NeurIPS Deep Inverse Workshop (poster)
9. **Kristina Monakhova**, Nick Antipa, Laura Waller, “Learning reconstructions for lensless imaging”, in 2019 Physics in ML Workshop, Berkeley, CA, May. 2019 (poster)
10. **Kristina Monakhova**, Kyrollos Yanny, Fanglin Linda Liu, Evan Shelhamer, Emrah Bostan, Laura Waller, “Deep Diffusers - machine learning for lensless imaging”, in 2018 ICCP Conference, Pittsburgh, PA, May. 2018 (poster)
11. Regina Eckert, **Kristina Monakhova**, Zachary F. Philips, Yongbing Zhang, Lei Tian, Laura Waller, “Advances in 3D Fourier Ptychography”, in 2017 ICCP Conference, Stanford, CA, May. 2017 (poster)

## Selected Invited Talks

---

|                                                                                                                                                        |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Toronto Vision Seminar</b>                                                                                                                          | fall 2024   |
| Talk: “Trustworthy and adaptive extreme low light imaging”                                                                                             |             |
| <b>CVPR UG2+ Workshop</b>                                                                                                                              | summer 2024 |
| <b>Janelia Computational Optics Conference</b>                                                                                                         | summer 2024 |
| Talk: “Learned, uncertainty-driven adaptive acquisition for scanning microscopy                                                                        |             |
| <b>Mitsubishi Electric Research Laboratories (MERL) Seminar Series</b>                                                                                 | fall 2023   |
| Talk: “Robust and physics-informed machine learning for low light imaging”                                                                             |             |
| <b>2023 Asilomar Conference on Signals, Systems, and Computers</b>                                                                                     | fall 2023   |
| Talks: “Physics-informed machine learning for computational imaging” and “Learned, uncertainty-driven adaptive acquisition for multiphoton microscopy” |             |
| <b>SPIE Optics + Photonics Emerging Topics in Artificial Intelligence</b>                                                                              | summer 2023 |
| Talk: “Physics-informed machine learning for computational imaging: From lensless cameras to low light videography”                                    |             |
| <b>Optica Imaging and Applied Optics Congress</b>                                                                                                      | summer 2023 |
| Talk: “Video Denoising in Starlight Using a Learned, Physics-informed Noise Model”                                                                     |             |
| <b>EPFL School of Computer and Communication Sciences Seminar</b>                                                                                      | spring 2023 |
| <b>Cornell Computer Science Seminar</b>                                                                                                                | spring 2023 |
| <b>University of Virginia ECE Seminar</b>                                                                                                              | spring 2023 |
| <b>Boston University ECE Seminar</b>                                                                                                                   | spring 2023 |
| <b>Carnegie Mellon University Meche Seminar</b>                                                                                                        | spring 2023 |
| Talk: “Cameras for better ML, and ML for better cameras”                                                                                               |             |
| <b>Cornell Artificial Intelligence Seminar</b>                                                                                                         | fall 2022   |
| Talk: “Physics-informed machine learning for computational imaging”                                                                                    |             |
| <b>MIT Media Lab Camera Culture Talk</b>                                                                                                               | summer 2022 |
| <b>Google Computational Imaging Workshop</b>                                                                                                           | summer 2022 |
| <b>Warren Grundfest Lectures in Computational Imaging</b>                                                                                              | summer 2022 |
| Talk: “Video denoising in starlight using a learned, physics-informed noise model”                                                                     |             |

|                                                                                                  |             |
|--------------------------------------------------------------------------------------------------|-------------|
| <b>CVPR Computational Cameras and Displays Workshop</b>                                          | summer 2022 |
| <b>Harvard Computational Imaging Seminar</b>                                                     | spring 2022 |
| Talk: “Physics-informed machine learning for compressive computational cameras”                  |             |
| <b>Berkeley Photobears Lightning Talk Series</b>                                                 | fall 2020   |
| Talk: “Compressive snapshot hyperspectral Imaging using a diffuser and a spectral filter array”  |             |
| <b>Berkeley Center for Computational Imaging Seminar Series</b>                                  | fall 2019   |
| Talk: “Practical mask-based lensless imaging reconstructions based on physics and deep learning” |             |
| <b>Berkeley Artificial Intelligence Research Lab Seminar Series</b>                              | fall 2019   |
| Talk: “Using physics and deep learning for practical imaging without a lens”                     |             |

## Advising

---

### GRADUATE STUDENTS AT CORNELL

|                              |                     |
|------------------------------|---------------------|
| Rawan Al Kubaisy (ECE PhD)   | spring 2025-present |
| Shamus Li (CS PhD)           | fall 2024-present   |
| Cassandra Ye (CS PhD)        | fall 2024-present   |
| Yujin (Haley) Lee (CS PhD)   | fall 2024-present   |
| Hasindu Kariyawasam (CS PhD) | fall 2024-present   |

### M.ENG STUDENTS AT CORNELL

|             |                       |
|-------------|-----------------------|
| Yumeng Jin  | spring 2025 - present |
| Syd Peng    | fall 2025             |
| Justin Kong | spring 2025           |

### UNDERGRADUATE STUDENTS AT CORNELL

|                                 |                       |
|---------------------------------|-----------------------|
| Nolan Lizmi                     | spring 2025 - present |
| Benjamin Li                     | fall 2025 - present   |
| Jonathan Zhang                  | fall 2025-present     |
| Shreya Majumdar (honors thesis) | fall 2025             |
| Adam Cahall                     | spring 2025           |

### DISSERTATION COMMITTEES AT CORNELL

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Peter Michael, CS PhD                              | fall 2024-present |
| Anusha Muralidharan, ECE PhD                       | fall 2024-present |
| Ziming Shao, Materials Science and Engineering PhD | fall 2024-2025    |
| Xiaoran Sui, Applied Physics PhD                   | fall 2024-present |

### EXTERNAL DISSERTATION COMMITTEES

|                                                                  |      |
|------------------------------------------------------------------|------|
| Eric Bezzam, EPFL, School of Computer and Communication Sciences | 2025 |
|------------------------------------------------------------------|------|

### UNDERGRADUATE AND GRADUATE STUDENTS AT UC BERKELEY AND MIT

|                                                                            |                    |
|----------------------------------------------------------------------------|--------------------|
| Cassandra Ye (now PhD student at Cornell)                                  | spring 2023 - 2024 |
| Christian Foley (currently at UC Berkeley)                                 | fall 2021 - 2024   |
| Shamus Li (now PhD student at Cornell)                                     | spring 2021 - 2022 |
| Mbalenhle Holt (BAIR REU, targeted at HBCUs)                               | summer 2021        |
| Georgia Channing (SUPERB REU, now at Global Computing Laboratory)          | summer 2021        |
| Yaying Zhao (UC Berkeley master's student, now at Facebook)                | summer 2020        |
| Vi Tran (Transfer to Excellence REU, now at Robinhood)                     | summer/fall 2020   |
| Trisha Sanghal (now a Software Engineer at Orchard)                        | 2019-2020          |
| Jonathan Fung (now at Scale AI)                                            | fall 2019          |
| Kristie Diep (BioESP REU, now at Medtronic)                                | summer 2019        |
| Ellin Zhao (now a PhD student at UCLA with Prof. Achuta Kadambi)           | 2018- 2020         |
| Joshua Yurtsever (now a Software Engineer at Google)                       | 2018-2020          |
| Nico Deshler (SUPERB REU, now a PhD student in Optics at Univ. of Arizona) | 2018-2020          |

## Service & Mentoring

---

### SERVICE AT CORNELL

|                                               |              |
|-----------------------------------------------|--------------|
| SWE Coffee Chat                               | 2026         |
| WICC bunch brunch speaker                     | 2025-present |
| TA training facilitator                       | sp 2026      |
| CS 6006 panelist                              | 2025         |
| PhD admissions committee, Graphics Area Chair | 2025         |

### SERVICE AT UC BERKELEY DURING PHD

|                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>UC Berkeley EECS Peer Mentor</b>                                                                                                                                                                                                                                                                                                                                                                                        | 2019-2022   |
| Held regular office hours to discuss issues and support junior PhD students                                                                                                                                                                                                                                                                                                                                                |             |
| <b>Berkeley Artificial Intelligence Research Mentoring (BAIR) Program</b>                                                                                                                                                                                                                                                                                                                                                  | 2018-2021   |
| Mentored undergraduate students from underrepresented groups interested in AI research                                                                                                                                                                                                                                                                                                                                     |             |
| <b>Equal Access for Application Assistance (EAAA) program</b>                                                                                                                                                                                                                                                                                                                                                              | fall 2021   |
| Volunteer application material reviewer to support diverse applicants to PhD programs                                                                                                                                                                                                                                                                                                                                      |             |
| <b>BAIR REU</b>                                                                                                                                                                                                                                                                                                                                                                                                            | summer 2021 |
| Research mentor for REU program for undergraduates from underrepresented backgrounds at HBCUs interested in AI research.                                                                                                                                                                                                                                                                                                   |             |
| <b>WICSE 1st year mentoring program</b>                                                                                                                                                                                                                                                                                                                                                                                    | 2017-2021   |
| Served as a mentor for 1st year female-identifying PhD students in the EECS Department                                                                                                                                                                                                                                                                                                                                     |             |
| <b>UC Berkeley Transfer-to-Excellence (TTE) REU</b>                                                                                                                                                                                                                                                                                                                                                                        | summer 2020 |
| Mentored an undergraduate researcher throughout summer REU program targeted at community college students coming from low-income backgrounds or underserved communities. Resulted in a journal paper for the student and the student successfully transferring to UC Berkeley.                                                                                                                                             |             |
| <b>EE Visit Days Coordinator</b>                                                                                                                                                                                                                                                                                                                                                                                           | spring 2020 |
| Organized the first Virtual Visit Days for admitted EECS PhD students. Organized peer advising program, matching all admitted students with a current graduate student mentor. Coordinated with underrepresented minority groups to hold virtual panels and discussions for admits. Coordinated student volunteers to promote casual admit-student interaction in a virtual setting, including virtual tours and hangouts. |             |
| <b>UC Berkeley EECS PhD Admissions Committee</b>                                                                                                                                                                                                                                                                                                                                                                           | winter 2020 |
| Reviewed PhD applications for the Signal Processing track in the EECS department.                                                                                                                                                                                                                                                                                                                                          |             |

|                                                                                                                                                                                                 |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>UC Berkeley Bioengineering Scholars Program (BioESP) Mentor</b>                                                                                                                              | summer 2019 |
| Mentored bioengineering undergraduate researcher throughout summer research program.                                                                                                            |             |
| <b>UC Berkeley SUPERB REU Mentor</b>                                                                                                                                                            | summer 2018 |
| Mentored undergraduate researcher during summer REU on a project involving thin, 3D cameras in array geometries. Student was selected to represent UC Berkeley at the 2018 REU Symposium.       |             |
| <b>Electrical Engineering Graduate Student Association</b>                                                                                                                                      | 2017-2018   |
| Served as social chair, worked to create inclusive and friendly environment for graduate students.                                                                                              |             |
| <b>Women in Computer Science and Electrical Engineering (WICSE)</b>                                                                                                                             | 2017-2018   |
| Organized events to promote diversity and inclusively within the EECS PhD program, including visit day events for female-identifying students, and mentorship program for 1st year PhD students |             |

## Professional Activities

---

### CONFERENCE SERVICE

|                                                                                                                       |      |
|-----------------------------------------------------------------------------------------------------------------------|------|
| Co-organizer, CVPR Computational Cameras and Displays (CCD) Workshop,<br>Nashville, TN, USA                           | 2025 |
| Poster and Demo Co-Chair, International Conference on Computational<br>Photography (ICCP) 2024, Lausanne, Switzerland | 2024 |
| Co-organizer, CVPR Computational Cameras and Displays (CCD) Workshop,<br>Seattle, WA, USA                             | 2024 |
| Poster and Demo Co-Chair, International Conference on Computational<br>Photography (ICCP) 2023, Madison, WI, USA      | 2023 |
| Co-organizer, AI/ML Theme, Optica Frontiers in Optics 2023, Tacoma, WA, USA                                           | 2023 |

### PROGRAM COMMITTEES

|                                                              |           |
|--------------------------------------------------------------|-----------|
| International Conference on Computational Photography (ICCP) | 2023-2025 |
| CVPR Computational Cameras and Displays Workshop             | 2021      |

### JOURNAL REVIEWING

IEEE Transactions on Pattern Analysis and Machine Intelligence  
 IEEE Transactions of Computational Imaging  
 Optica - Continuum, Optics Letters, Optics Express  
 ACM Computing Surveys  
 Science Advances

### EXTERNAL SERVICE

|                                                                            |              |
|----------------------------------------------------------------------------|--------------|
| Advisory Board, University at Buffalo Department of Electrical Engineering | 2026-present |
|----------------------------------------------------------------------------|--------------|